

16.11 Global Closure and System-Level Guarantees (Final — Clean Version)

Building upon the components developed in Sections 16.1–16.10, we establish the global closure and system-level guarantees of the SEKHEM-TSDD framework.

16.11.1 Unified System State

$$\mathcal{S}_k = (\Psi_k, P_k, PAC_k, L_k, D_k, u_k)$$

$$\mathcal{S}_k \in \mathcal{X}_{\{\mathrm{safe}\}}$$

where $\mathcal{X}_{\{\mathrm{safe}\}}$ denotes the admissible safe state space.

16.11.2 Global Evolution Law

$$\mathcal{S}^{k+1} = F_{\{\mathrm{global}\}}(\mathcal{S}_k)$$

with:

$$\begin{aligned}
&F_{\{\mathrm{global}\}} = \\
&F_{\{\mathrm{dyn}\}} \circ \\
&F_{\{\mathrm{prob}\}} \circ \\
&F_{\{\mathrm{PAC}\}} \circ \\
&F_{\{\mathrm{graph}\}} \circ \\
&F_{\{\mathrm{decision}\}} \circ \\
&F_{\{\mathrm{control}\}}
\end{aligned}$$

Thus:

the system evolves through a fully integrated composition of all subsystems.

16.11.3 Closed-Loop Structure

$$\begin{aligned}
&\Psi \rightarrow P \rightarrow PAC \rightarrow L \\
&\rightarrow D \rightarrow u \rightarrow \Psi
\end{aligned}$$

Thus:

the system is fully self-contained with no external dependency.

16.11.4 Feasibility Invariance

$\mathcal{S}_k \in \mathcal{X}_{\{\text{safe}\}}$
 $\rightarrow$
 $\mathcal{S}_{k+1} \in \mathcal{X}_{\{\text{safe}\}}$

Thus:

the system remains within the safe domain for all time.

16.11.5 Global Lyapunov Stability

$\mathcal{L}(\mathcal{S}) =$
 $\mathcal{E}(\Psi) + KL(P \parallel P_0)$

$\frac{d\mathcal{L}}{dt}$
 $\leq$
 $-c \|\nabla \mathcal{E}(\Psi)\|^2$
 $+ C\sigma^2$

Thus:

global stability is preserved under stochastic perturbations.

16.11.6 Robustness

Assuming bounded noise, delay, and disturbances:

$$\|\mathcal{S}_k\| \leq C$$

Thus:

the system is globally bounded (ISS-type robustness).

16.11.7 Convergence

$$\lim_{k \rightarrow \infty} \mathcal{S}_k = \mathcal{S}^*$$

Thus:

the system converges to a steady-state solution under standard regularity conditions.

16.11.8 Optimality

$$u_k = \arg\max_{u \in \mathcal{U}_{\mathrm{feasible}}} D_u(k)$$

Thus:

the system implements an optimal decision policy.

16.11.9 Adaptivity

- PAC evolves dynamically
- network topology adapts
- control law updates in real time

Thus:

the system is inherently self-adaptive.

16.11.10 Real-Time Guarantee

$$T_{\mathrm{total}} \leq T_{\mathrm{critical}}$$

Thus:

the system satisfies real-time constraints.

16.11.11 Safety Closure

- state constraints enforced
- fail-safe mechanisms active
- bounded trajectories ensured

$\Rightarrow \mathcal{S}_k \in \mathcal{X}_{\{\mathrm{safe}\}}$

16.11.12 Multi-Agent Consistency

$\lambda_2(L) > 0$

Thus:

the system achieves consensus across agents.

16.11.13 Deterministic Limit

$\sigma \rightarrow 0$
 $\Rightarrow$
 $\mathcal{S}_k \rightarrow \mathcal{S}^*$

Thus:

the system reduces to a deterministic optimal solution.

Final Master Statement

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\text{SEKHEM-TSDD defines a fully closed,
stochastic, triadic decision–control system with
guaranteed stability, safety, optimality, and
adaptivity.}
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SEKHEM-TSDD Theory, Mohamed Abd Elnaby,
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